

fong-spivak-invitation: image evidence

Image regions

| Identifier                      | Page | Conf. | LaTeX source | Rendered | Image                                                                                                                                                                                                                                                                                                                                                                                                         |
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| fong-spivak-invitation_DIA_0001 | 12   | —     | —            | —        | <div>Prepare lemon meringue pie</div> <div><div>prepared crust</div><div>lemon</div><div>butter</div><div>sugar</div><div>egg</div><div>separate egg</div><div>yolk</div><div>white</div><div>sugar</div><div>make lemon filling</div><div>make meringue</div><div>fill crust</div><div>lemon filling</div><div>unbaked lemon pie</div><div>add meringue</div><div>meringue</div><div>unbaked pie</div></div> |
| fong-spivak-invitation_DIA_0002 | 16   | —     | —            | —        |                                                                                                                                                                                                                                                                                                                                                                                                               |
| fong-spivak-invitation_DIA_0003 | 17   | —     | —            | —        |                                                                                                                                                                                                                                                                                                                                                                                                               |
| fong-spivak-invitation_DIA_0004 | 17   | —     | —            | —        |                                                                                                                                                                                                                                                                                                                                                                                                               |
| fong-spivak-invitation_DIA_0005 | 17   | —     | —            | —        |                                                                                                                                                                                                                                                                                                                                                                                                               |
| fong-spivak-invitation_DIA_0006 | 17   | —     | —            | —        |                                                                                                                                                                                                                                                                                                                                                                                                               |
| fong-spivak-invitation_DIA_0007 | 17   | —     | —            | —        |                                                                                                                                                                                                                                                                                                                                                                                                               |
| fong-spivak-invitation_DIA_0008 | 17   | —     | —            | —        |                                                                                                                                                                                                                                                                                                                                                                                                               |

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| fong-spivak-invitation_<br>DIA_0009 | 17   | —     | —            | —        |                    |
| fong-spivak-invitation_<br>DIA_0010 | 18   | —     | —            | —        |                    |
| fong-spivak-invitation_<br>DIA_0011 | 18   | —     | —            | —        |                    |
| fong-spivak-invitation_<br>DIA_0012 | 18   | —     | —            | —        |                    |
| fong-spivak-invitation_<br>DIA_0013 | 19   | —     | —            | —        |                    |
| fong-spivak-invitation_<br>DIA_0014 | 19   | —     | —            | —        |                    |
| fong-spivak-invitation_<br>DIA_0015 | 20   | —     | —            | —        | true<br>↑<br>false |
| fong-spivak-invitation_<br>DIA_0016 | 23   | —     | —            | —        |                    |

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| fong-spivak-invitation_DIA_0017 | 25   | —     | —            | —        | <div> <div>arbitrary function</div> <div><math>\underline{3} \rightarrow \underline{3}</math></div> </div> <div> <div>surjective function</div> <div><math>\underline{3} \twoheadrightarrow \underline{2}</math></div> </div> <div> <div>injective function</div> <div><math>\underline{2} \hookrightarrow \underline{3}</math></div> </div> <div> <div>bijective function</div> <div><math>\underline{3} \xrightarrow{\cong} \underline{3}</math></div> </div> |
| fong-spivak-invitation_DIA_0018 | 25   | —     | —            | —        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| fong-spivak-invitation_DIA_0019 | 25   | —     | —            | —        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| fong-spivak-invitation_DIA_0020 | 25   | —     | —            | —        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| fong-spivak-invitation_DIA_0021 | 25   | —     | —            | —        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| fong-spivak-invitation_DIA_0022 | 25   | —     | —            | —        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| fong-spivak-invitation_DIA_0023 | 26   | —     | —            | —        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| fong-spivak-invitation_DIA_0024 | 26   | —     | —            | —        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| fong-spivak-invitation_DIA_0025 | 26   | —     | —            | —        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| fong-spivak-invitation_DIA_0026 | 27   | —     | —            | —        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| fong-spivak-invitation_DIA_0027 | 27   | —     | —            | —        | <div> <div>true</div> <div>↑</div> <div>false</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                      |

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| fong-spivak-invitation_DIA_0028 | 28   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0029 | 30   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0030 | 30   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0031 | 30   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0032 | 31   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0033 | 31   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0034 | 32   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0035 | 33   | —     | —            | —        |       |

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| fong-spivak-invitation_<br>DIA_0036 | 33   | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0037 | 33   | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0038 | 36   | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0039 | 36   | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0040 | 36   | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0041 | 39   | —     | —            | —        |       |

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| fong-spivak-invitation_DIA_0042 | 39   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0043 | 42   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0044 | 42   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0045 | 43   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0046 | 43   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0047 | 43   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0048 | 44   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0049 | 46   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0050 | 50   | —     | —            | —        |       |

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| fong-spivak-invitation_DIA_0051 | 53   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0052 | 53   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0053 | 56   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0054 | 56   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0055 | 56   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0056 | 56   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0057 | 56   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0058 | 56   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0059 | 57   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0060 | 57   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0061 | 57   | —     | —            | —        |       |

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| fong-spivak-invitation_DIA_0062 | 57   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0063 | 58   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0064 | 58   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0065 | 58   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0066 | 58   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0067 | 58   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0068 | 59   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0069 | 59   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0070 | 59   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0071 | 59   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0072 | 59   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0073 | 60   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0074 | 62   | —     | —            | —        |       |

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| fong-spivak-invitation_DIA_0075 | 63   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0076 | 63   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0077 | 64   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0078 | 64   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0079 | 64   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0080 | 64   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0081 | 70   | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0082 | 73   | —     | —            | —        |       |

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| fong-spivak-invitation_<br>DIA_0083 | 75   | —     | —            | —        | $X :=$                             |
| fong-spivak-invitation_<br>DIA_0084 | 75   | —     | —            | —        | $=: Y$                             |
| fong-spivak-invitation_<br>DIA_0085 | 80   | —     | —            | —        | $\mathcal{X} :=$                   |
| fong-spivak-invitation_<br>DIA_0086 | 80   | —     | —            | —        | $=: \mathcal{Y}$                   |
| fong-spivak-invitation_<br>DIA_0087 | 81   | —     | —            | —        | $\mathcal{X} \times \mathcal{Y} =$ |
| fong-spivak-invitation_<br>DIA_0088 | 88   | —     | —            | —        | $Y :=$                             |
| fong-spivak-invitation_<br>DIA_0089 | 92   | —     | —            | —        | $\text{easySchema} :=$             |

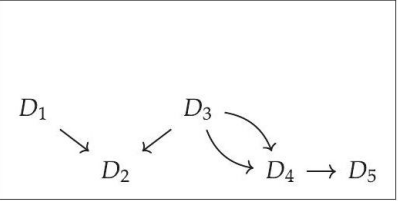
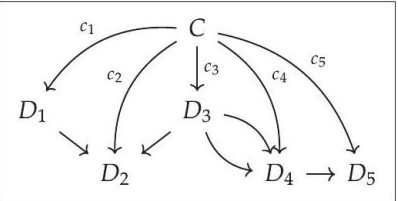
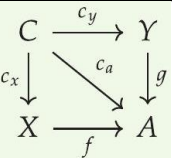
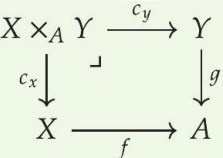
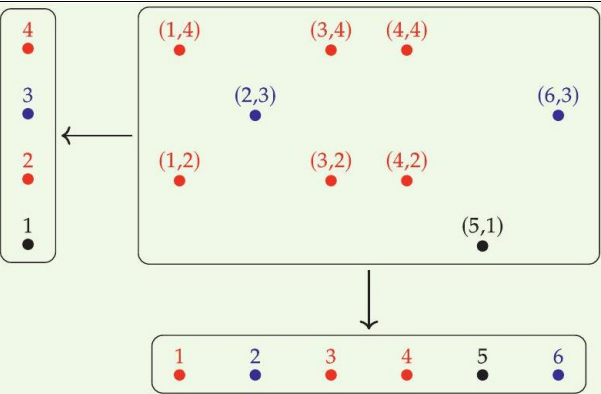
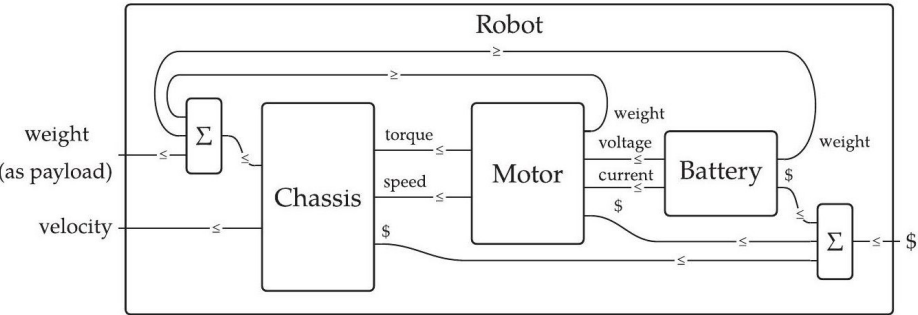
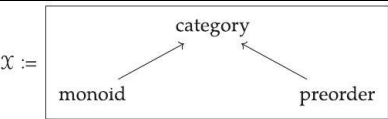
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| fong-spivak-invitation_<br>DIA_0090 | 93   | —     | —            | —        | <div> <div> <math display="block">  \begin{array}{ccccc}  &amp; \text{Mngr} &amp; &amp; \text{Employee} &amp; &amp; \text{Department} \\  &amp; \curvearrowright &amp; &amp; \bullet &amp; \xleftrightarrow{\text{WorksIn}} &amp; \bullet \\  &amp; &amp; &amp; \searrow &amp; \xleftarrow{\text{Secr}} &amp; \nearrow \\  &amp; &amp; &amp; \text{FName} &amp; &amp; \text{DName} \\  &amp; &amp; &amp; &amp; \text{string} &amp; \\  &amp; &amp; &amp; &amp; \circ &amp;   \end{array}  </math> </div> <div> <math display="block">  \begin{array}{l}  \text{Department.Secr.WorksIn} = \text{Department} \\  \text{Employee.Mngr.WorksIn} = \text{Employee.WorksIn}  \end{array}  </math> </div> </div>                               |
| fong-spivak-invitation_<br>DIA_0091 | 93   | —     | —            | —        | <pre> schema mySchema = {   nodes     Employee, Department;   attributes     DName : Department -&gt; string,     FName : Employee   -&gt; string;   arrows     Mngr   : Employee   -&gt; Employee,     WorksIn : Employee -&gt; Department,     Secr   : Department -&gt; Employee;   equations     Department.Secr.WorksIn = Department,     Employee.Mngr.WorksIn   = Employee.WorksIn; } </pre>                                                                                                                                                                                                                                                                                                                                      |
| fong-spivak-invitation_<br>DIA_0092 | 94   | —     | —            | —        | <div> <div> <math display="block">  \begin{array}{ccccc}  &amp; &amp; &amp; \\$ &amp; \\  &amp; &amp; &amp; \circ &amp; \\  &amp; \nearrow \text{Price} &amp; &amp; \nwarrow \text{Price} &amp; \\  \text{Economy} &amp; &amp; &amp; &amp; \text{First Class} \\  \bullet &amp; &amp; &amp; &amp; \bullet \\  \searrow \text{Position} &amp; &amp; &amp; \swarrow \text{Position} &amp; \\  &amp; &amp; &amp; \text{string} &amp; \\  &amp; &amp; &amp; \circ &amp;   \end{array}  </math> </div> <div> <math display="block">  \begin{array}{c}  \\$ \\  \circ \\  \uparrow \text{Price} \\  \text{Airline Seat} \\  \bullet \\  \downarrow \text{Position} \\  \text{string} \\  \circ  \end{array}  \quad =: B  </math> </div> </div> |
| fong-spivak-invitation_<br>DIA_0093 | 97   | —     | —            | —        | <div> <math display="block">  \begin{array}{c}  s \\  \curvearrowright \\  \bullet \\  z  \end{array}  </math> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| fong-spivak-invitation_<br>DIA_0094 | 98   | —     | —            | —        | <div> <div> <math display="block">  \begin{array}{ccc}  A &amp; \xrightarrow{f} &amp; B \\  \bullet &amp; &amp; \bullet \\  g \downarrow &amp; &amp; \downarrow h \\  C &amp; \xrightarrow{i} &amp; D \\  \bullet &amp; &amp; \bullet  \end{array}  </math> <p><i>no equations</i></p> </div> <div> <math display="block">  \begin{array}{ccc}  A &amp; \xrightarrow{f} &amp; B \\  \bullet &amp; &amp; \bullet \\  g \downarrow &amp; &amp; \downarrow h \\  C &amp; \xrightarrow{i} &amp; D \\  \bullet &amp; &amp; \bullet  \end{array}  </math> <p><math>f \circ h = g \circ i</math></p> </div> </div>                                                                                                                              |
| fong-spivak-invitation_<br>DIA_0095 | 98   | —     | —            | —        | <div> <math display="block">  \begin{array}{ccc}  A &amp; \xrightarrow{f} &amp; B \\  \bullet &amp; &amp; \bullet \\  g \downarrow &amp; \searrow j &amp; \downarrow h \\  C &amp; \xrightarrow{i} &amp; D \\  \bullet &amp; &amp; \bullet  \end{array}  </math> <p><math>f \circ h = j \circ g \circ i</math></p> </div>                                                                                                                                                                                                                                                                                                                                                                                                                |

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| fong-spivak-invitation_<br>DIA_0096 | 99   | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0097 | 100  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0098 | 100  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0099 | 100  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0100 | 100  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0101 | 103  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0102 | 103  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0103 | 104  | —     | —            | —        |       |

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| fong-spivak-invitation_DIA_0104 | 104  | —     | —            | —        | <div> <div>Prime number</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| fong-spivak-invitation_DIA_0105 | 104  | —     | —            | —        | <div> <div>Flying pig</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| fong-spivak-invitation_DIA_0106 | 105  | —     | —            | —        | <div> <div> Beatle → Rock-and-roll instrument </div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| fong-spivak-invitation_DIA_0107 | 105  | —     | —            | —        | <div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| fong-spivak-invitation_DIA_0108 | 105  | —     | —            | —        | <div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| fong-spivak-invitation_DIA_0109 | 105  | —     | —            | —        | <div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| fong-spivak-invitation_DIA_0110 | 106  | —     | —            | —        | <div> <div>Free_square :=</div> <div> <math display="block">  \begin{array}{ccc}  A' &amp; \xrightarrow{f'} &amp; B' \\  g' \downarrow &amp; &amp; \downarrow h' \\  C' &amp; \xrightarrow{i'} &amp; D'  \end{array}  </math> </div> <div>no equations</div> <div>Comm_square :=</div> <div> <math display="block">  \begin{array}{ccc}  A &amp; \xrightarrow{f} &amp; B \\  g \downarrow &amp; &amp; \downarrow h \\  C &amp; \xrightarrow{i} &amp; D  \end{array}  </math> </div> <div><math>f \circ h = g \circ i</math></div> </div> |
| fong-spivak-invitation_DIA_0111 | 109  | —     | —            | —        | <div> <div> <math display="block">  \begin{array}{c}  s \\  \curvearrowright \\  \bullet \\  z  \end{array}  </math> </div> <div><math>s \circ s = z</math></div> </div>                                                                                                                                                                                                                                                                                                                                                                 |
| fong-spivak-invitation_DIA_0112 | 109  | —     | —            | —        | <div> <div> <math display="block">  \begin{array}{ccccc}  a &amp; \xrightarrow{f} &amp; b &amp; \xrightarrow[g]{h} &amp; c \\  &amp; &amp; &amp; &amp; \\  &amp; &amp; f \circ g = f \circ h &amp; &amp;   \end{array}  </math> </div> </div>                                                                                                                                                                                                                                                                                            |

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| fong-spivak-invitation_<br>DIA_0113 | 109  | —     | —            | —        |                                                                                                                                                                                                                                                                                                             |
| fong-spivak-invitation_<br>DIA_0114 | 110  | —     | —            | —        | $\begin{array}{ccc} F(c) & \xrightarrow{\alpha_c} & G(c) \\ F(f) \downarrow & & \downarrow G(f) \\ F(d) & \xrightarrow{\alpha_d} & G(d) \end{array}$                                                                                                                                                        |
| fong-spivak-invitation_<br>DIA_0115 | 110  | —     | —            | —        |                                                                                                                                                                                                                                                                                                             |
| fong-spivak-invitation_<br>DIA_0116 | 114  | —     | —            | —        | $\begin{array}{ccc} G(\text{Arrow}) & \xrightarrow{\alpha_{\text{Arrow}}} & H(\text{Arrow}) \\ G(\text{source}) \downarrow & & \downarrow H(\text{source}) \\ G(\text{Vertex}) & \xrightarrow{\alpha_{\text{Vertex}}} & H(\text{Vertex}) \end{array}$                                                       |
| fong-spivak-invitation_<br>DIA_0117 | 114  | —     | —            | —        | $\begin{array}{ccc} G(\text{Arrow}) & \xrightarrow{\alpha_{\text{Arrow}}} & H(\text{Arrow}) \\ G(\text{target}) \downarrow & & \downarrow H(\text{target}) \\ G(\text{Vertex}) & \xrightarrow{\alpha_{\text{Vertex}}} & H(\text{Vertex}) \end{array}$                                                       |
| fong-spivak-invitation_<br>DIA_0118 | 115  | —     | —            | —        | $\text{Gr} := \begin{array}{ccc} \text{Arrow} & \xrightleftharpoons[\text{target}]{\text{source}} & \text{Vertex} \\ \bullet & & \bullet \\ \text{no equations} \end{array} \quad \text{DDS} := \begin{array}{c} \text{State} \\ \bullet \\ \updownarrow \\ \text{next} \\ \text{no equations} \end{array}$ |
| fong-spivak-invitation_<br>DIA_0119 | 116  | —     | —            | —        |                                                                                                                                                                                                                                                                                                             |
| fong-spivak-invitation_<br>DIA_0120 | 116  | —     | —            | —        |                                                                                                                                                                                                                                                                                                             |
| fong-spivak-invitation_<br>DIA_0121 | 117  | —     | —            | —        | $\mathcal{C} \xrightarrow{F} \mathcal{D} \begin{array}{c} \xrightarrow{I} \\ \Downarrow \alpha \\ \xrightarrow{J} \end{array} \text{Set}$                                                                                                                                                                   |

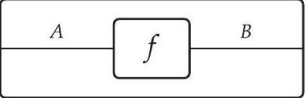
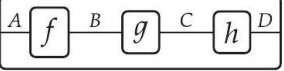
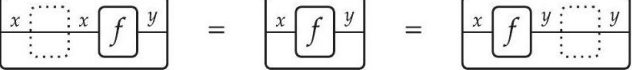
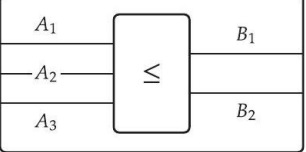
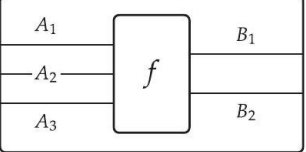
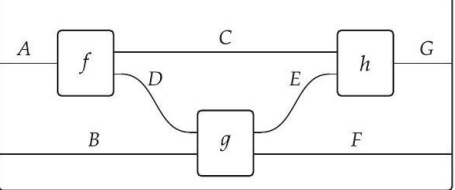
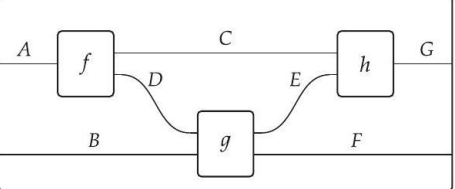
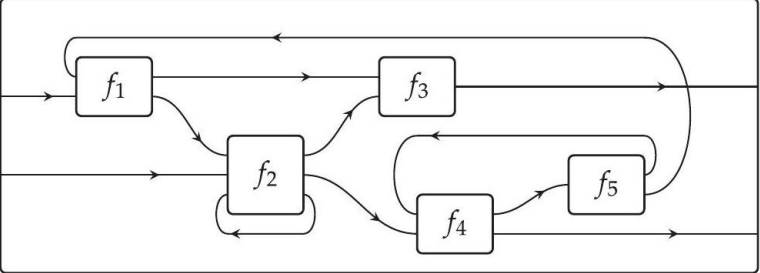
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|---------------------------------|------|-------|--------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fong-spivak-invitation_DIA_0122 | 117  | —     | —            | —        |                                                                                                                                                                                                                                                        |
| fong-spivak-invitation_DIA_0123 | 118  | —     | —            | —        | $\begin{array}{ccc} \mathcal{C}(c, Rd) & \xrightarrow{\alpha_{c,d}} & \mathcal{D}(Lc, d) \\ \mathcal{C}(f, Rg) \downarrow & & \downarrow \mathcal{D}(Lf, g) \\ \mathcal{C}(c', Rd') & \xrightarrow{\alpha_{c',d'}} & \mathcal{D}(Lc', d') \end{array}$ |
| fong-spivak-invitation_DIA_0124 | 119  | —     | —            | —        |                                                                                                                                                                                                                                                        |
| fong-spivak-invitation_DIA_0125 | 120  | —     | —            | —        |                                                                                                                                                                                                                                                        |
| fong-spivak-invitation_DIA_0126 | 124  | —     | —            | —        |                                                                                                                                                                                                                                                        |
| fong-spivak-invitation_DIA_0127 | 124  | —     | —            | —        |                                                                                                                                                                                                                                                        |
| fong-spivak-invitation_DIA_0128 | 125  | —     | —            | —        |                                                                                                                                                                                                                                                        |
| fong-spivak-invitation_DIA_0129 | 126  | —     | —            | —        |                                                                                                                                                                                                                                                        |

| Identifier                      | Page | Conf. | LaTeX source | Rendered | Image                                                                                 |
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| fong-spivak-invitation_DIA_0130 | 126  | —     | —            | —        |    |
| fong-spivak-invitation_DIA_0131 | 126  | —     | —            | —        |    |
| fong-spivak-invitation_DIA_0132 | 128  | —     | —            | —        |    |
| fong-spivak-invitation_DIA_0133 | 128  | —     | —            | —        |    |
| fong-spivak-invitation_DIA_0134 | 128  | —     | —            | —        |   |
| fong-spivak-invitation_DIA_0135 | 132  | —     | —            | —        |  |
| fong-spivak-invitation_DIA_0136 | 134  | —     | —            | —        |  |

| Identifier                          | Page | Conf. | LaTeX source | Rendered | Image                                                                                                             |
|-------------------------------------|------|-------|--------------|----------|-------------------------------------------------------------------------------------------------------------------|
| fong-spivak-invitation_<br>DIA_0137 | 134  | —     | —            | —        | $\mathcal{Y} := \begin{array}{c} \text{this book} \\ \uparrow \\ \text{nothing} \end{array}$                      |
| fong-spivak-invitation_<br>DIA_0138 | 136  | —     | —            | —        |                                                                                                                   |
| fong-spivak-invitation_<br>DIA_0139 | 137  | —     | —            | —        |                                                                                                                   |
| fong-spivak-invitation_<br>DIA_0140 | 138  | —     | —            | —        |                                                                                                                   |
| fong-spivak-invitation_<br>DIA_0141 | 138  | —     | —            | —        |                                                                                                                   |
| fong-spivak-invitation_<br>DIA_0142 | 138  | —     | —            | —        | $T := \begin{array}{c} \text{good-natured} \\ \bullet \\ \uparrow \\ \text{mean-spirited} \\ \bullet \end{array}$ |
| fong-spivak-invitation_<br>DIA_0143 | 138  | —     | —            | —        | $E := \begin{array}{c} \text{funny} \\ \bullet \\ \uparrow \\ \text{boring} \\ \bullet \end{array}$               |

| Identifier                          | Page | Conf. | LaTeX source | Rendered | Image |
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| fong-spivak-invitation_<br>DIA_0144 | 138  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0145 | 139  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0146 | 140  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0147 | 141  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0148 | 142  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0149 | 143  | —     | —            | —        |       |

| Identifier                          | Page | Conf. | LaTeX source | Rendered | Image |
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| fong-spivak-invitation_<br>DIA_0150 | 143  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0151 | 144  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0152 | 146  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0153 | 146  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0154 | 147  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0155 | 148  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0156 | 148  | —     | —            | —        |       |

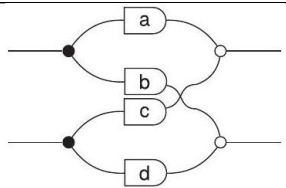
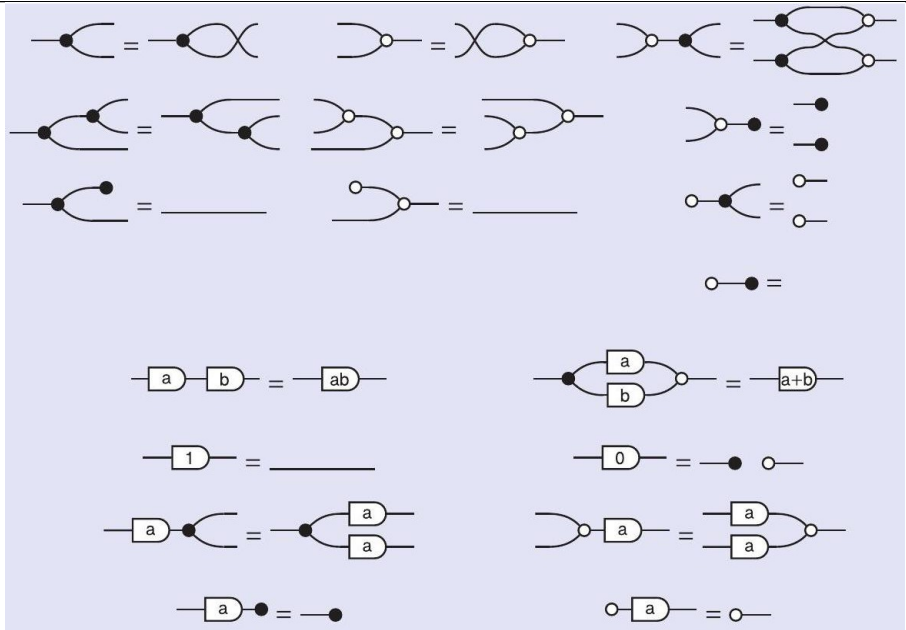
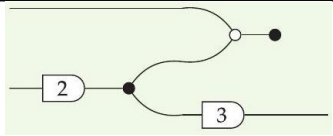
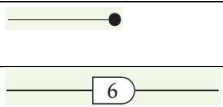
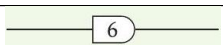
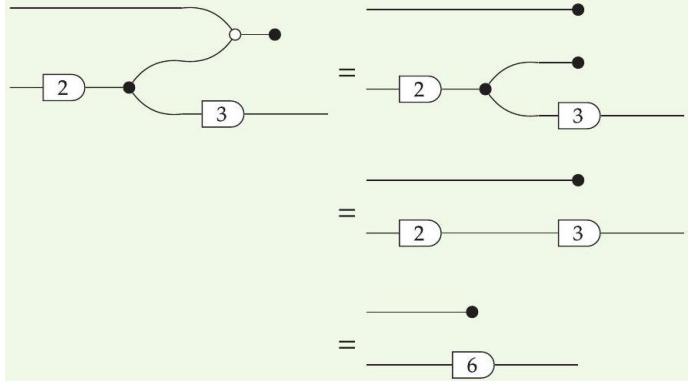
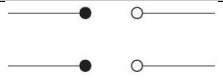
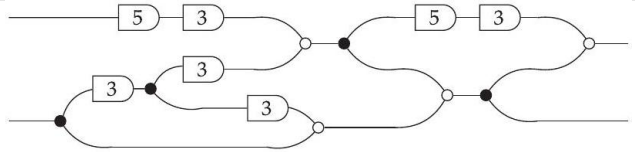
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| fong-spivak-invitation_DIA_0157 | 148  | —     | —            | —        |    |
| fong-spivak-invitation_DIA_0158 | 148  | —     | —            | —        |    |
| fong-spivak-invitation_DIA_0159 | 149  | —     | —            | —        |    |
| fong-spivak-invitation_DIA_0160 | 149  | —     | —            | —        |    |
| fong-spivak-invitation_DIA_0161 | 149  | —     | —            | —        |    |
| fong-spivak-invitation_DIA_0162 | 149  | —     | —            | —        |    |
| fong-spivak-invitation_DIA_0163 | 149  | —     | —            | —        |    |
| fong-spivak-invitation_DIA_0164 | 149  | —     | —            | —        |   |
| fong-spivak-invitation_DIA_0165 | 149  | —     | —            | —        |  |
| fong-spivak-invitation_DIA_0166 | 152  | —     | —            | —        |  |
| fong-spivak-invitation_DIA_0167 | 154  | —     | —            | —        |  |

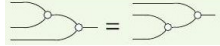
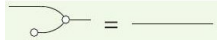
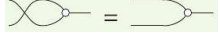
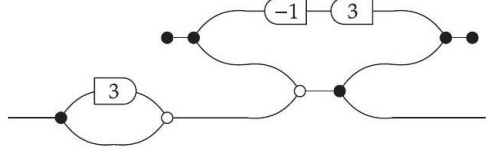
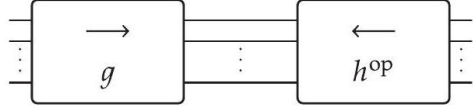
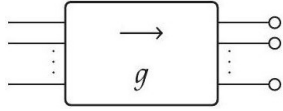
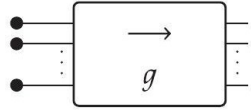
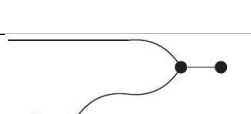
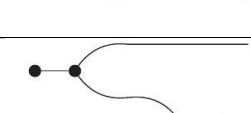
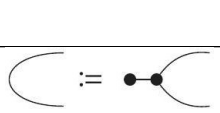
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| fong-spivak-invitation_<br>DIA_0168 | 154  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0169 | 154  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0170 | 155  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0171 | 155  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0172 | 155  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0173 | 155  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0174 | 156  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0175 | 156  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0176 | 156  | —     | —            | —        |       |

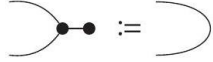
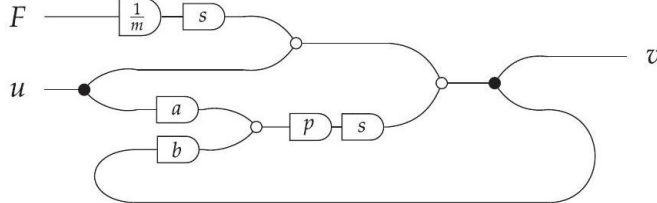
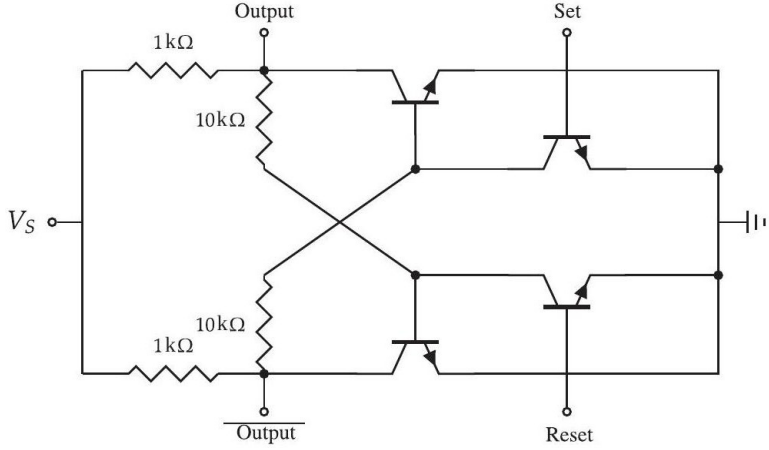
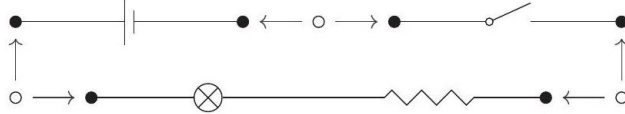
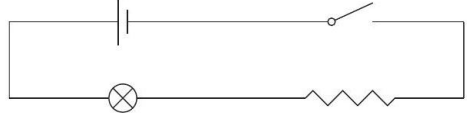
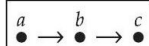
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| fong-spivak-invitation_DIA_0177 | 157  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0178 | 157  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0179 | 158  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0180 | 161  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0181 | 162  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0182 | 164  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0183 | 165  | —     | —            | —        |       |

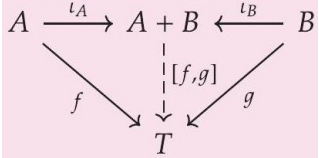
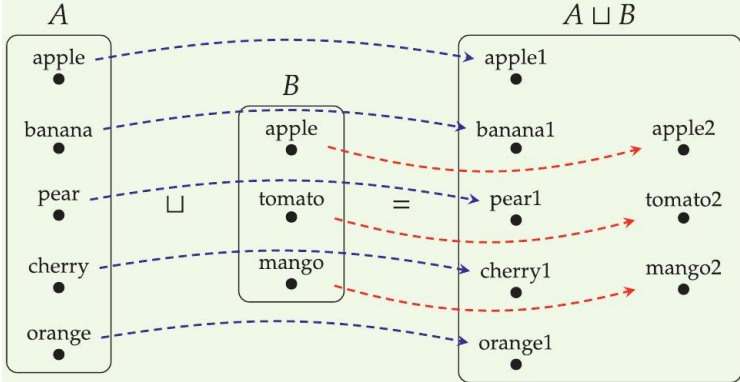
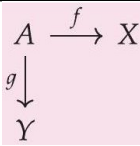
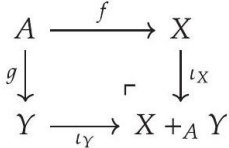
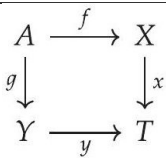
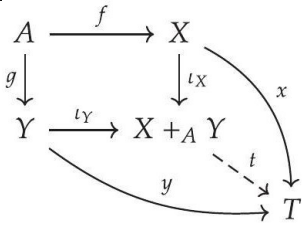
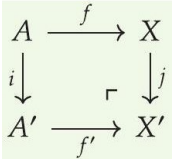
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| fong-spivak-invitation_DIA_0184 | 166  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0185 | 166  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0186 | 167  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0187 | 170  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0188 | 172  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0189 | 174  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0190 | 175  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0191 | 175  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0192 | 175  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0193 | 175  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0194 | 175  | —     | —            | —        |       |

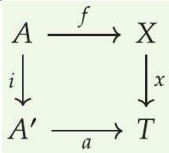
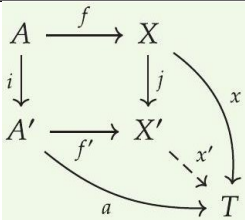
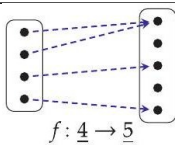
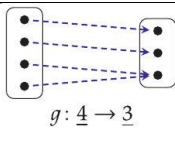
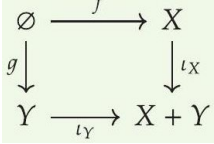
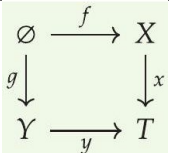
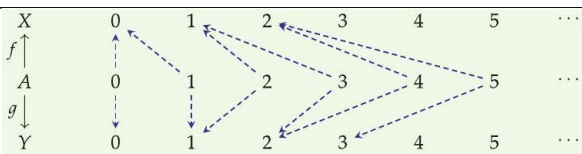
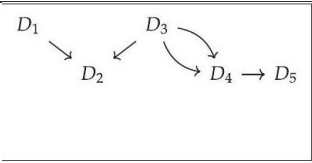
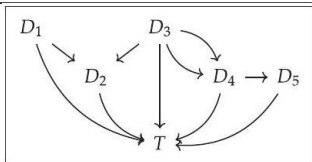
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| fong-spivak-invitation_<br>DIA_0196 | 176  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0197 | 176  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0198 | 177  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0199 | 177  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0200 | 177  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0201 | 179  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0202 | 181  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0203 | 181  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0204 | 181  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0205 | 182  | —     | —            | —        |       |

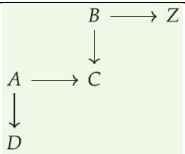
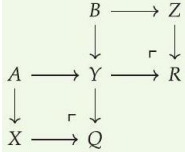
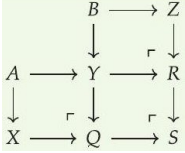
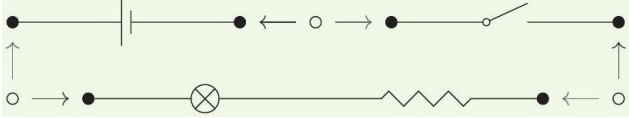
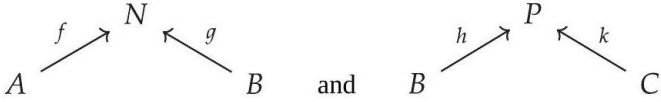
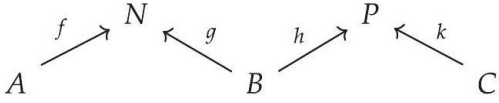
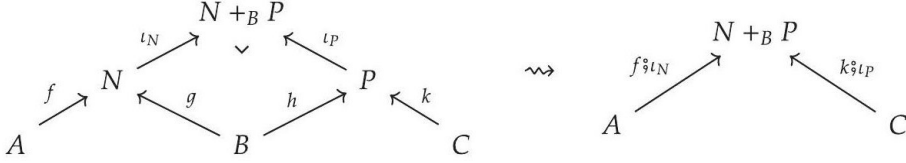
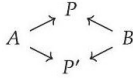
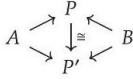
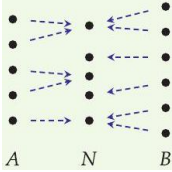
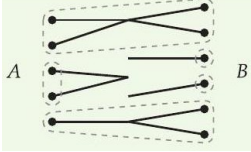
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| fong-spivak-invitation_<br>DIA_0207 | 184  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0208 | 185  | —     | —            | —        |   |
| fong-spivak-invitation_<br>DIA_0209 | 185  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0210 | 185  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0211 | 185  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0212 | 185  | —     | —            | —        |  |
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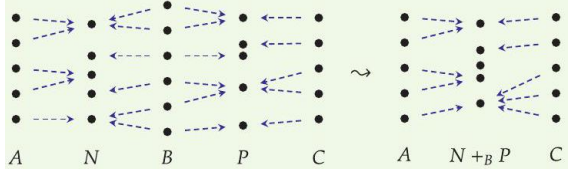
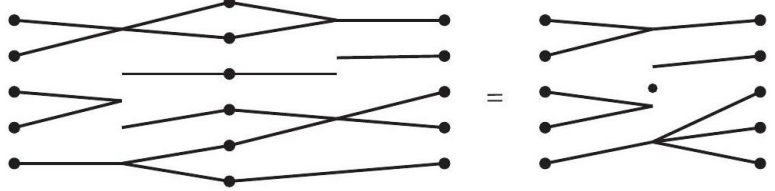
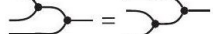
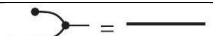
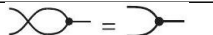
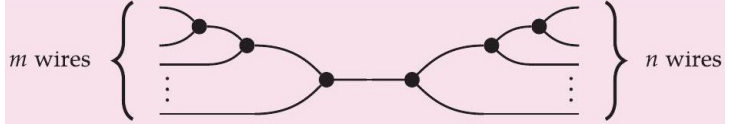
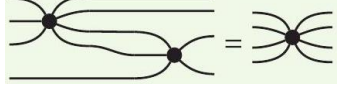
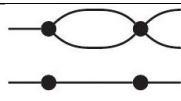
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| fong-spivak-invitation_DIA_0215 | 186  | —     | —            | —        |    |
| fong-spivak-invitation_DIA_0216 | 186  | —     | —            | —        |    |
| fong-spivak-invitation_DIA_0217 | 188  | —     | —            | —        |    |
| fong-spivak-invitation_DIA_0218 | 189  | —     | —            | —        |    |
| fong-spivak-invitation_DIA_0219 | 190  | —     | —            | —        |    |
| fong-spivak-invitation_DIA_0220 | 190  | —     | —            | —        |    |
| fong-spivak-invitation_DIA_0221 | 190  | —     | —            | —        |   |
| fong-spivak-invitation_DIA_0222 | 191  | —     | —            | —        |  |
| fong-spivak-invitation_DIA_0223 | 191  | —     | —            | —        |  |
| fong-spivak-invitation_DIA_0224 | 191  | —     | —            | —        |  |
| fong-spivak-invitation_DIA_0225 | 192  | —     | —            | —        |  |
| fong-spivak-invitation_DIA_0226 | 192  | —     | —            | —        |  |
| fong-spivak-invitation_DIA_0227 | 192  | —     | —            | —        |  |
| fong-spivak-invitation_DIA_0228 | 192  | —     | —            | —        |  |

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| fong-spivak-invitation_<br>DIA_0229 | 192  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0230 | 192  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0231 | 194  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0232 | 195  | —     | —            | —        |   |
| fong-spivak-invitation_<br>DIA_0233 | 195  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0234 | 195  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0235 | 195  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0236 | 196  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0237 | 196  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0238 | 198  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0239 | 198  | —     | —            | —        |  |

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| fong-spivak-invitation_<br>DIA_0241 | 198  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0242 | 199  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0243 | 200  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0244 | 201  | —     | —            | —        |   |
| fong-spivak-invitation_<br>DIA_0245 | 201  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0246 | 201  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0247 | 202  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0248 | 202  | —     | —            | —        |  |

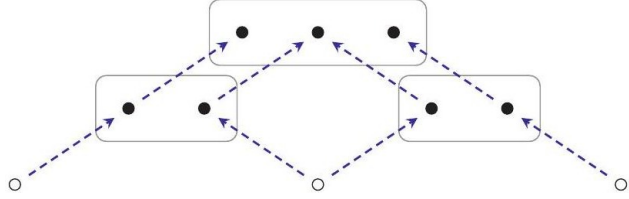
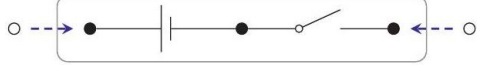
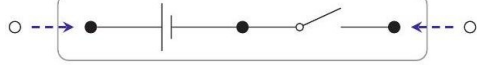
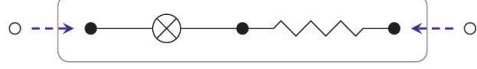
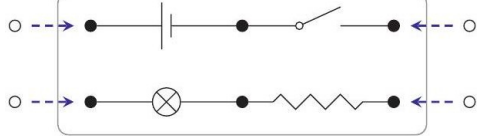
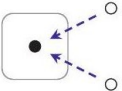
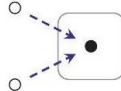
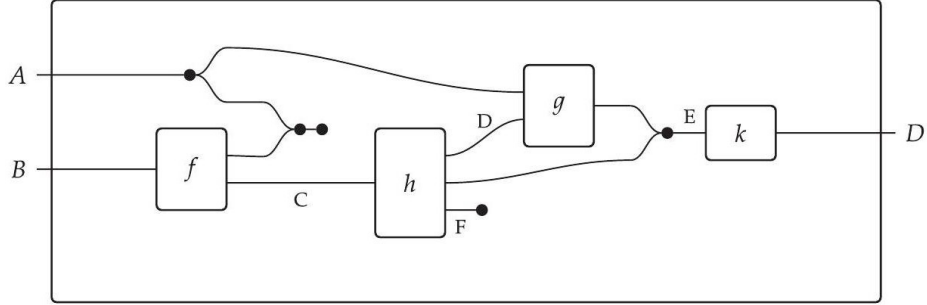
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| fong-spivak-invitation_<br>DIA_0250 | 202  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0251 | 203  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0252 | 203  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0253 | 203  | —     | —            | —        |   |
| fong-spivak-invitation_<br>DIA_0254 | 203  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0255 | 204  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0256 | 204  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0257 | 204  | —     | —            | —        |  |

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| fong-spivak-invitation_DIA_0258 | 205  | —     | —            | —        |    |
| fong-spivak-invitation_DIA_0259 | 205  | —     | —            | —        |    |
| fong-spivak-invitation_DIA_0260 | 205  | —     | —            | —        |    |
| fong-spivak-invitation_DIA_0261 | 207  | —     | —            | —        |    |
| fong-spivak-invitation_DIA_0262 | 208  | —     | —            | —        |    |
| fong-spivak-invitation_DIA_0263 | 208  | —     | —            | —        |   |
| fong-spivak-invitation_DIA_0264 | 208  | —     | —            | —        |  |
| fong-spivak-invitation_DIA_0265 | 208  | —     | —            | —        |  |
| fong-spivak-invitation_DIA_0266 | 208  | —     | —            | —        |  |
| fong-spivak-invitation_DIA_0267 | 209  | —     | —            | —        |  |
| fong-spivak-invitation_DIA_0268 | 209  | —     | —            | —        |  |

| Identifier                          | Page | Conf. | LaTeX source | Rendered | Image                                                                                 |
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| fong-spivak-invitation_<br>DIA_0269 | 210  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0270 | 210  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0271 | 211  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0272 | 211  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0273 | 211  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0274 | 211  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0275 | 211  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0276 | 211  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0277 | 211  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0278 | 212  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0279 | 212  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0280 | 212  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0281 | 213  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0282 | 213  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0283 | 213  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0284 | 213  | —     | —            | —        |  |

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| fong-spivak-invitation_DIA_0285 | 213  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0286 | 213  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0287 | 214  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0288 | 215  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0289 | 216  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0290 | 216  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0291 | 216  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0292 | 217  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0293 | 218  | —     | —            | —        |       |

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| fong-spivak-invitation_<br>DIA_0294 | 218  | —     | —            | —        |                                                                                                              |
| fong-spivak-invitation_<br>DIA_0295 | 218  | —     | —            | —        |                                                                                                              |
| fong-spivak-invitation_<br>DIA_0296 | 218  | —     | —            | —        |                                                                                                              |
| fong-spivak-invitation_<br>DIA_0297 | 219  | —     | —            | —        |                                                                                                              |
| fong-spivak-invitation_<br>DIA_0298 | 220  | —     | —            | —        |                                                                                                              |
| fong-spivak-invitation_<br>DIA_0299 | 221  | —     | —            | —        |                                                                                                              |
| fong-spivak-invitation_<br>DIA_0300 | 221  | —     | —            | —        |                                                                                                              |
| fong-spivak-invitation_<br>DIA_0301 | 221  | —     | —            | —        | $b := \bullet \text{---}     \text{---} \bullet$ and $s := \bullet \text{---} \text{---} \text{---} \bullet$ |
| fong-spivak-invitation_<br>DIA_0302 | 222  | —     | —            | —        |                                                                                                              |
| fong-spivak-invitation_<br>DIA_0303 | 222  | —     | —            | —        |                                                                                                              |
| fong-spivak-invitation_<br>DIA_0304 | 222  | —     | —            | —        |                                                                                                              |
| fong-spivak-invitation_<br>DIA_0305 | 222  | —     | —            | —        |                                                                                                              |

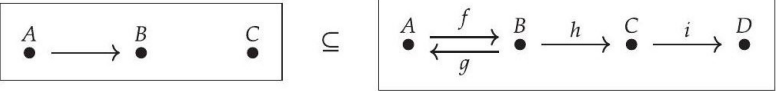
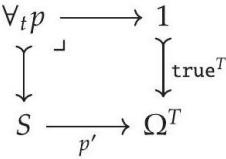
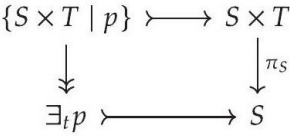
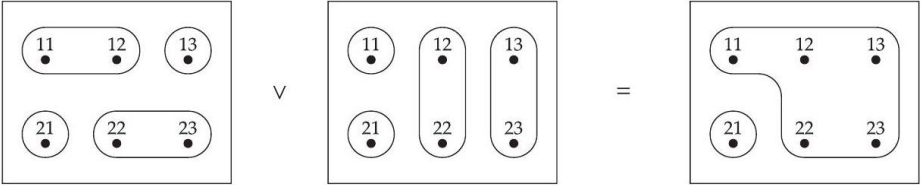
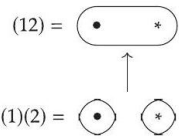
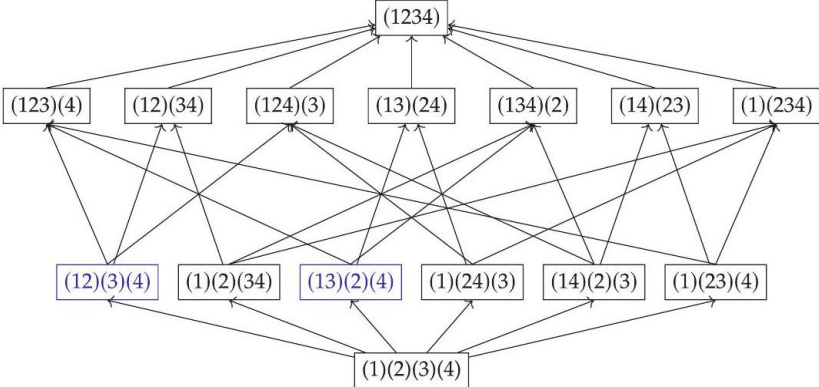
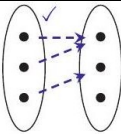
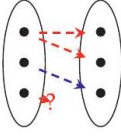
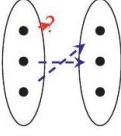
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| fong-spivak-invitation_<br>DIA_0306 | 222  | —     | —            | —        |                            |
| fong-spivak-invitation_<br>DIA_0307 | 223  | —     | —            | —        |                            |
| fong-spivak-invitation_<br>DIA_0308 | 223  | —     | —            | —        |                            |
| fong-spivak-invitation_<br>DIA_0309 | 223  | —     | —            | —        |                            |
| fong-spivak-invitation_<br>DIA_0310 | 223  | —     | —            | —        |                            |
| fong-spivak-invitation_<br>DIA_0311 | 223  | —     | —            | —        |                            |
| fong-spivak-invitation_<br>DIA_0312 | 224  | —     | —            | —        |                            |
| fong-spivak-invitation_<br>DIA_0313 | 224  | —     | —            | —        |                           |
| fong-spivak-invitation_<br>DIA_0314 | 224  | —     | —            | —        |                          |
| fong-spivak-invitation_<br>DIA_0315 | 224  | —     | —            | —        | $\eta := \emptyset$      |
| fong-spivak-invitation_<br>DIA_0316 | 224  | —     | —            | —        |  $\emptyset =: \epsilon$ |
| fong-spivak-invitation_<br>DIA_0317 | 225  | —     | —            | —        |                          |

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| fong-spivak-invitation_<br>DIA_0318 | 226  | —     | —            | —        |              |
| fong-spivak-invitation_<br>DIA_0319 | 227  | —     | —            | —        |              |
| fong-spivak-invitation_<br>DIA_0320 | 228  | —     | —            | —        |              |
| fong-spivak-invitation_<br>DIA_0321 | 229  | —     | —            | —        |              |
| fong-spivak-invitation_<br>DIA_0322 | 229  | —     | —            | —        |              |
| fong-spivak-invitation_<br>DIA_0323 | 231  | —     | —            | —        | $\varphi :=$ |
| fong-spivak-invitation_<br>DIA_0324 | 231  | —     | —            | —        |              |
| fong-spivak-invitation_<br>DIA_0325 | 231  | —     | —            | —        |              |

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| fong-spivak-invitation_<br>DIA_0326 | 233  | —     | —            | —        |                                                                                                                                                                                                                                                                                                           |
| fong-spivak-invitation_<br>DIA_0327 | 234  | —     | —            | —        |                                                                                                                                                                                                                                                                                                           |
| fong-spivak-invitation_<br>DIA_0328 | 235  | —     | —            | —        |                                                                                                                                                                                                                                                                                                           |
| fong-spivak-invitation_<br>DIA_0329 | 237  | —     | —            | —        | $\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow & & \downarrow \lrcorner & & \downarrow \\ D & \longrightarrow & E & \longrightarrow & F \end{array}$                                                                                                                   |
| fong-spivak-invitation_<br>DIA_0330 | 237  | —     | —            | —        | $\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow \lrcorner & & \downarrow & & \downarrow \\ D & \longrightarrow & E & \longrightarrow & F \end{array}$                                                                                                                   |
| fong-spivak-invitation_<br>DIA_0331 | 237  | —     | —            | —        | $\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow \lrcorner & & \downarrow \lrcorner & & \downarrow \\ A' & \longrightarrow & B' & \longrightarrow & C' \end{array}$                                                                                                      |
| fong-spivak-invitation_<br>DIA_0332 | 237  | —     | —            | —        | $\begin{array}{ccc} A & \xrightarrow{\text{id}_A} & A \\ \text{id}_A \downarrow \lrcorner & & \downarrow f \\ A & \xrightarrow{f} & B \end{array} \quad \begin{array}{ccc} A & \xrightarrow{f} & B \\ f \downarrow & \lrcorner & \downarrow \text{id}_B \\ B & \xrightarrow{\text{id}_B} & B \end{array}$ |
| fong-spivak-invitation_<br>DIA_0333 | 238  | —     | —            | —        | $\begin{array}{ccc} A' & \xrightarrow{f'} & B' \\ i' \downarrow \cong \lrcorner & & \cong \downarrow i \\ A & \xrightarrow{f} & B \end{array}$                                                                                                                                                            |
| fong-spivak-invitation_<br>DIA_0334 | 238  | —     | —            | —        | $\begin{array}{ccc} A & \xrightarrow{f} & B \\ \parallel \lrcorner & & \parallel \\ A & \xrightarrow{f} & B \end{array}$                                                                                                                                                                                  |
| fong-spivak-invitation_<br>DIA_0335 | 238  | —     | —            | —        | $\begin{array}{ccc} A' & \xrightarrow{g} & A \\ f' \downarrow \lrcorner & & \downarrow f \\ B' & \xrightarrow{h} & B \end{array}$                                                                                                                                                                         |
| fong-spivak-invitation_<br>DIA_0336 | 239  | —     | —            | —        |                                                                                                                                                                                                                                                                                                           |

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| fong-spivak-invitation_<br>DIA_0337 | 240  | —     | —            | —        |                                                                                                                                                                                   |
| fong-spivak-invitation_<br>DIA_0338 | 240  | —     | —            | —        |                                                                                                                                                                                   |
| fong-spivak-invitation_<br>DIA_0339 | 245  | —     | —            | —        |                                                                                                                                                                                   |
| fong-spivak-invitation_<br>DIA_0340 | 245  | —     | —            | —        |                                                                                                                                                                                   |
| fong-spivak-invitation_<br>DIA_0341 | 245  | —     | —            | —        |                                                                                                                                                                                   |
| fong-spivak-invitation_<br>DIA_0342 | 245  | —     | —            | —        |                                                                                                                                                                                   |
| fong-spivak-invitation_<br>DIA_0343 | 247  | —     | —            | —        | <p>An open set <math>U \subseteq \mathbb{R}^2</math>, a point <math>p = (x, y) \in U</math>,<br/>and an <math>\epsilon</math>-ball <math>B(x, y; \epsilon) \subseteq U</math></p> |
| fong-spivak-invitation_<br>DIA_0344 | 250  | —     | —            | —        |                                                                                                                                                                                   |

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| fong-spivak-invitation_DIA_0345 | 251  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0346 | 251  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0347 | 252  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0348 | 252  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0349 | 253  | —     | —            | —        |       |
| fong-spivak-invitation_DIA_0350 | 257  | —     | —            | —        |       |

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| fong-spivak-invitation_<br>DIA_0351 | 257  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0352 | 262  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0353 | 262  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0354 | 270  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0355 | 270  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0356 | 271  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0357 | 272  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0358 | 272  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0359 | 272  | —     | —            | —        |  |

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| fong-spivak-invitation_<br>DIA_0360 | 272  | —     | —            | —        |                                                                                                               |
| fong-spivak-invitation_<br>DIA_0361 | 273  | —     | —            | —        | <p> <math>\bullet \mapsto p_1</math><br/> <math>* \mapsto p_2</math><br/> <math>\circ \mapsto p_3</math> </p> |
| fong-spivak-invitation_<br>DIA_0362 | 274  | —     | —            | —        |                                                                                                               |
| fong-spivak-invitation_<br>DIA_0363 | 274  | —     | —            | —        | $\emptyset$                                                                                                   |
| fong-spivak-invitation_<br>DIA_0364 | 274  | —     | —            | —        | $\{1\}$<br>$\uparrow$<br>$\emptyset$                                                                          |
| fong-spivak-invitation_<br>DIA_0365 | 274  | —     | —            | —        | $\{1,2\}$<br>$\swarrow \quad \searrow$<br>$\{1\} \quad \{2\}$<br>$\swarrow \quad \searrow$<br>$\emptyset$     |
| fong-spivak-invitation_<br>DIA_0366 | 275  | —     | —            | —        |                                                                                                               |
| fong-spivak-invitation_<br>DIA_0367 | 275  | —     | —            | —        |                                                                                                               |

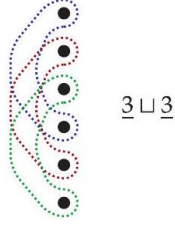
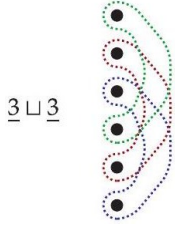
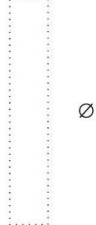
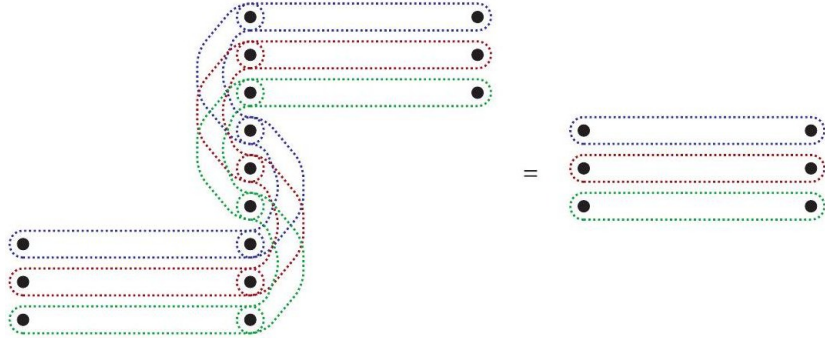
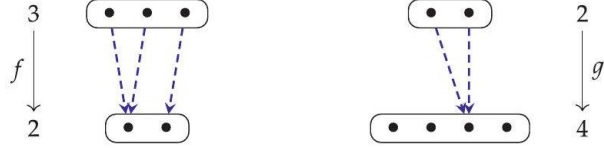
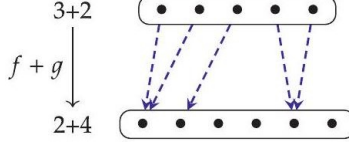
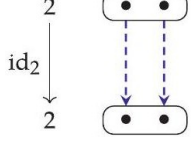
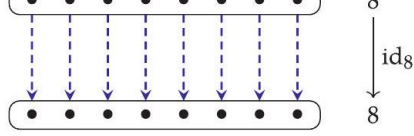
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| fong-spivak-invitation_<br>DIA_0368 | 275  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0369 | 275  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0370 | 276  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0371 | 278  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0372 | 278  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0373 | 278  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0374 | 278  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0375 | 278  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0376 | 278  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0377 | 278  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0378 | 278  | —     | —            | —        |       |

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| fong-spivak-invitation_DIA_0379 | 278  | —     | —            | —        | $g!(c_3) = \begin{array}{c} 12 \quad 3 \quad 4 \\ \swarrow \quad \searrow \quad \downarrow \\ \bullet \quad \bullet \end{array}$                                                                                                                                                                                                                                                             |
| fong-spivak-invitation_DIA_0380 | 278  | —     | —            | —        | $g!(c_4) = \begin{array}{c} 12 \quad 3 \quad 4 \\ \swarrow \quad \searrow \quad \downarrow \\ \bullet \quad \bullet \end{array}$                                                                                                                                                                                                                                                             |
| fong-spivak-invitation_DIA_0381 | 279  | —     | —            | —        | $\begin{array}{c} 1 \quad 2 \quad 3 \quad 4 \\ \swarrow \quad \searrow \quad \swarrow \quad \searrow \\ \bullet \end{array}$                                                                                                                                                                                                                                                                 |
| fong-spivak-invitation_DIA_0382 | 279  | —     | —            | —        | $\begin{array}{c} 1 \quad 2 \quad 3 \quad 4 \\ \swarrow \quad \searrow \quad \swarrow \quad \searrow \\ \bullet \quad \bullet \end{array}$                                                                                                                                                                                                                                                   |
| fong-spivak-invitation_DIA_0383 | 279  | —     | —            | —        | $\begin{array}{c} 1 \quad 2 \quad 3 \quad 4 \\ \swarrow \quad \searrow \quad \swarrow \quad \searrow \\ \bullet \quad \bullet \end{array}$                                                                                                                                                                                                                                                   |
| fong-spivak-invitation_DIA_0384 | 279  | —     | —            | —        | $\begin{array}{c} 1 \quad 2 \quad 3 \quad 4 \\ \swarrow \quad \searrow \quad \swarrow \quad \searrow \\ \bullet \quad \bullet \end{array}$                                                                                                                                                                                                                                                   |
| fong-spivak-invitation_DIA_0385 | 279  | —     | —            | —        | $\begin{array}{c} 1 \quad 2 \quad 3 \quad 4 \\ \swarrow \quad \searrow \quad \swarrow \quad \searrow \\ \bullet \quad \bullet \quad \bullet \end{array}$                                                                                                                                                                                                                                     |
| fong-spivak-invitation_DIA_0386 | 279  | —     | —            | —        | $c = \begin{array}{c} 1 \quad 2 \quad 3 \quad 4 \\ \swarrow \quad \searrow \quad \swarrow \quad \searrow \\ \bullet \quad \bullet \end{array}$                                                                                                                                                                                                                                               |
| fong-spivak-invitation_DIA_0387 | 279  | —     | —            | —        | $g!(c) = \begin{array}{c} 12 \quad 3 \quad 4 \\ \swarrow \quad \searrow \quad \downarrow \\ \bullet \quad \bullet \end{array}$                                                                                                                                                                                                                                                               |
| fong-spivak-invitation_DIA_0388 | 279  | —     | —            | —        | $d = \begin{array}{c} 12 \quad 3 \quad 4 \\ \swarrow \quad \searrow \quad \swarrow \quad \searrow \\ \bullet \end{array}$                                                                                                                                                                                                                                                                    |
| fong-spivak-invitation_DIA_0389 | 279  | —     | —            | —        | $e = \begin{array}{c} 12 \quad 3 \quad 4 \\ \swarrow \quad \searrow \quad \swarrow \quad \searrow \\ \bullet \quad \bullet \end{array}$                                                                                                                                                                                                                                                      |
| fong-spivak-invitation_DIA_0390 | 279  | —     | —            | —        | $g^*(d) = \begin{array}{c} 1 \quad 2 \quad 3 \quad 4 \\ \swarrow \quad \searrow \quad \swarrow \quad \searrow \\ \bullet \end{array}$                                                                                                                                                                                                                                                        |
| fong-spivak-invitation_DIA_0391 | 279  | —     | —            | —        | $g^*(e) = \begin{array}{c} 1 \quad 2 \quad 3 \quad 4 \\ \swarrow \quad \searrow \quad \swarrow \quad \searrow \\ \bullet \quad \bullet \end{array}$                                                                                                                                                                                                                                          |
| fong-spivak-invitation_DIA_0392 | 280  | —     | —            | —        | $P := \begin{array}{c} \boxed{\begin{array}{c} 4 \\ \bullet \\ \uparrow \\ 3.9 \\ \bullet \\ \swarrow \quad \searrow \\ 1 \quad 2 \\ \bullet \quad \bullet \end{array}} \begin{array}{c} \xleftarrow{g} \\ \xrightarrow{f} \end{array} \begin{array}{c} \boxed{\begin{array}{c} 4 \\ \bullet \\ \swarrow \quad \searrow \\ 1 \quad 2 \\ \bullet \quad \bullet \end{array}} =: Q \end{array}$ |

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| fong-spivak-invitation_<br>DIA_0393 | 281  | —     | —            | —        | <p> <math>X := \begin{array}{c} \bullet \\ a_1 \end{array} \quad \begin{array}{c} \bullet \\ c_2 \\ \bullet \\ c_1 \end{array}</math><br/> <math>f \downarrow</math><br/> <math>Y := \begin{array}{c} \bullet \\ a \end{array} \quad \begin{array}{c} \bullet \\ b \end{array} \quad \begin{array}{c} \bullet \\ c \end{array}</math> </p> |
| fong-spivak-invitation_<br>DIA_0394 | 281  | —     | —            | —        |                                                                                                                                                                                                                                                                                                                                            |
| fong-spivak-invitation_<br>DIA_0395 | 285  | —     | —            | —        |                                                                                                                                                                                                                                                                                                                                            |
| fong-spivak-invitation_<br>DIA_0396 | 286  | —     | —            | —        |                                                                                                                                                                                                                                                                                                                                            |
| fong-spivak-invitation_<br>DIA_0397 | 286  | —     | —            | —        |                                                                                                                                                                                                                                                                                                                                            |
| fong-spivak-invitation_<br>DIA_0398 | 291  | —     | —            | —        | <p>Free_square :=</p> <p>no equations</p>                                                                                                                                                                                                                                                                                                  |
| fong-spivak-invitation_<br>DIA_0399 | 292  | —     | —            | —        |                                                                                                                                                                                                                                                                                                                                            |

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| fong-spivak-invitation_<br>DIA_0400 | 292  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0401 | 292  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0402 | 294  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0403 | 294  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0404 | 294  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0405 | 294  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0406 | 295  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0407 | 295  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0408 | 296  | —     | —            | —        |       |

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| fong-spivak-invitation_<br>DIA_0409 | 296  | —     | —            | —        | <p>Diagram illustrating a directed edge (arrow) between two vertices. The arrow is labeled 'source' at the tail and 'target' at the head. The vertices are labeled 'Vertex'.</p>                                                                                                                                                                                                                                                                                                                                                        |
| fong-spivak-invitation_<br>DIA_0410 | 296  | —     | —            | —        | <p>Diagram illustrating a directed graph with nodes labeled 1 through 7. Edges are labeled with numbers: 1, 2, 3, 4, 5, 6, 7. The graph shows a cycle 1 → 2 → 3 → 4 → 5 → 6 → 7 → 1, with additional self-loops on nodes 1, 4, and 5.</p>                                                                                                                                                                                                                                                                                               |
| fong-spivak-invitation_<br>DIA_0411 | 297  | —     | —            | —        | <p>Diagram illustrating a directed graph with nodes labeled E, B, G, P, S, D. Edges are labeled with numbers: 1, 2, 3, 4, 5, 6. The graph shows a cycle E → B → G → P → S → D → E, with additional self-loops on nodes B and S.</p>                                                                                                                                                                                                                                                                                                     |
| fong-spivak-invitation_<br>DIA_0412 | 299  | —     | —            | —        | <p>Diagram illustrating a directed graph with nodes labeled (monoid, this book), (preorder, this book), (category, this book), (monoid, nothing), (preorder, nothing), (category, nothing). Edges are labeled with numbers: 1, 2, 3, 4, 5, 6. The graph shows a cycle (monoid, nothing) → (category, nothing) → (preorder, nothing) → (category, this book) → (preorder, this book) → (monoid, this book) → (monoid, nothing), with additional self-loops on nodes (monoid, nothing), (category, nothing), and (preorder, nothing).</p> |
| fong-spivak-invitation_<br>DIA_0413 | 302  | —     | —            | —        | <p>Diagram illustrating a directed graph with nodes labeled A, B, C, D. Edges are labeled with numbers: 1, 2, 3, 4, 5, 6. The graph shows a cycle A → B → C → D → A, with additional self-loops on nodes A, B, C, and D.</p>                                                                                                                                                                                                                                                                                                            |
| fong-spivak-invitation_<br>DIA_0414 | 303  | —     | —            | —        | <p>Diagram illustrating a directed graph with nodes labeled A, B, C, D, x, y, z. Edges are labeled with numbers: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11. The graph shows a cycle A → B → C → D → A, with additional self-loops on nodes A, B, C, and D, and edges between x, y, z.</p>                                                                                                                                                                                                                                                       |
| fong-spivak-invitation_<br>DIA_0415 | 304  | —     | —            | —        | <p>Diagram illustrating a directed graph with nodes labeled A, B, C, D. Edges are labeled with numbers: 1, 2, 3, 4, 5, 6. The graph shows a cycle A → B → C → D → A, with additional self-loops on nodes A, B, C, and D.</p>                                                                                                                                                                                                                                                                                                            |

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| fong-spivak-invitation_<br>DIA_0416 | 304  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0417 | 304  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0418 | 304  | —     | —            | —        |    |
| fong-spivak-invitation_<br>DIA_0419 | 305  | —     | —            | —        |   |
| fong-spivak-invitation_<br>DIA_0420 | 306  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0421 | 306  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0422 | 306  | —     | —            | —        |  |
| fong-spivak-invitation_<br>DIA_0423 | 306  | —     | —            | —        |  |

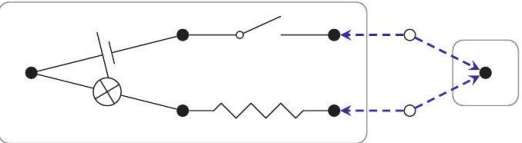
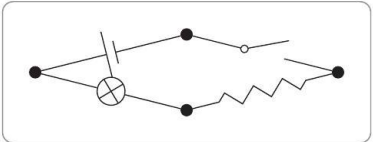
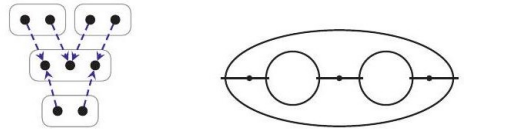
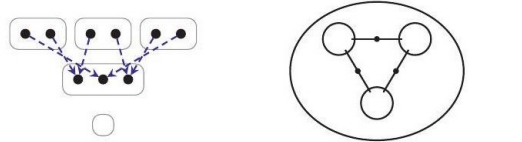
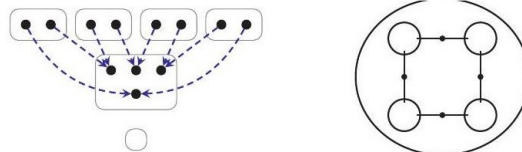
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| fong-spivak-invitation_<br>DIA_0424 | 307  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0425 | 308  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0426 | 308  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0427 | 308  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0428 | 309  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0429 | 309  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0430 | 311  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0431 | 312  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0432 | 312  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0433 | 312  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0434 | 312  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0435 | 312  | —     | —            | —        |       |

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| fong-spivak-invitation_<br>DIA_0436 | 312  | —     | —            | —        | $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$      |
| fong-spivak-invitation_<br>DIA_0437 | 312  | —     | —            | —        |                                                             |
| fong-spivak-invitation_<br>DIA_0438 | 313  | —     | —            | —        | $\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \sim$            |
| fong-spivak-invitation_<br>DIA_0439 | 313  | —     | —            | —        | $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \sim$         |
| fong-spivak-invitation_<br>DIA_0440 | 313  | —     | —            | —        | $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \sim$ |
| fong-spivak-invitation_<br>DIA_0441 | 314  | —     | —            | —        |                                                             |
| fong-spivak-invitation_<br>DIA_0442 | 314  | —     | —            | —        |                                                             |
| fong-spivak-invitation_<br>DIA_0443 | 314  | —     | —            | —        |                                                             |
| fong-spivak-invitation_<br>DIA_0444 | 314  | —     | —            | —        |                                                             |
| fong-spivak-invitation_<br>DIA_0445 | 314  | —     | —            | —        |                                                             |
| fong-spivak-invitation_<br>DIA_0446 | 314  | —     | —            | —        | $\begin{pmatrix} 0 \end{pmatrix} =$                         |

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| fong-spivak-invitation_DIA_0447 | 314  | —     | —            | —        |                                                                                                                                                                                                                                                       |
| fong-spivak-invitation_DIA_0448 | 315  | —     | —            | —        | $  \begin{array}{ccc}  (a, b, c) & \xrightarrow{(\mu \otimes \text{id})} & (a * b, c) \\  (\text{id} \otimes \mu) \downarrow & & \downarrow \mu \\  (a, b * c) & \xrightarrow{\mu} & a * b * c  \end{array}  $                                        |
| fong-spivak-invitation_DIA_0449 | 315  | —     | —            | —        | $  \begin{array}{ccc}  a & \xrightarrow{(\eta \otimes \text{id})} & (1, a) \\  (\text{id} \otimes \eta) \downarrow & \searrow \text{id}_a & \downarrow \mu \\  (a, 1) & \xrightarrow{\mu} & a  \end{array}  $                                         |
| fong-spivak-invitation_DIA_0450 | 315  | —     | —            | —        | $  \begin{array}{ccc}  (a, b) & \xrightarrow{\sigma} & (b, a) \\  \swarrow \mu & & \downarrow \mu \\  & & a * b  \end{array}  $                                                                                                                       |
| fong-spivak-invitation_DIA_0451 | 315  | —     | —            | —        | $  \begin{array}{ccc}  (a, b, c) & \xrightarrow{(\mu \otimes \text{id})} & (a + b, c) \\  (\text{id} \otimes \mu) \downarrow & & \downarrow \mu \\  (a, b + c) & \xrightarrow{\mu} & a + b + c  \end{array}  $                                        |
| fong-spivak-invitation_DIA_0452 | 315  | —     | —            | —        | $  \begin{array}{ccc}  a & \xrightarrow{(\eta \otimes \text{id})} & (0, a) \\  (\text{id} \otimes \eta) \downarrow & \searrow \text{id}_a & \downarrow \mu \\  (a, 0) & \xrightarrow{\mu} & a  \end{array}  $                                         |
| fong-spivak-invitation_DIA_0453 | 315  | —     | —            | —        | $  \begin{array}{ccc}  (a, b) & \xrightarrow{\sigma} & (b, a) \\  \swarrow \mu & & \downarrow \mu \\  & & a + b  \end{array}  $                                                                                                                       |
| fong-spivak-invitation_DIA_0454 | 317  | —     | —            | —        |                                                                                                                                                                                                                                                       |
| fong-spivak-invitation_DIA_0455 | 317  | —     | —            | —        |                                                                                                                                                                                                                                                       |
| fong-spivak-invitation_DIA_0456 | 317  | —     | —            | —        |                                                                                                                                                                                                                                                       |
| fong-spivak-invitation_DIA_0457 | 319  | —     | —            | —        | $  \begin{array}{ccccc}  A & \xrightarrow{\iota_A} & A + B & \xleftarrow{\iota_B} & B \\  & \searrow f & \downarrow [f, g] & \swarrow g & \\  & & C & & \\  f \circ h \swarrow & & \downarrow h & & \searrow g \circ h \\  & & D & &   \end{array}  $ |
| fong-spivak-invitation_DIA_0458 | 319  | —     | —            | —        | $  \begin{array}{ccccc}  A & \xrightarrow{\iota_A} & A + B & \xleftarrow{\iota_B} & B \\  & \searrow \iota_A & \parallel \text{id}_{A+B} & \swarrow \iota_B & \\  & & A + B & &   \end{array}  $                                                      |

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| fong-spivak-invitation_<br>DIA_0459 | 320  | —     | —            | —        | $ \begin{array}{ccccc} A & \xrightarrow{\iota_A} & A+B & \xleftarrow{\iota_B} & B \\ & \searrow f & \downarrow f+g & \swarrow g & \\ C & \xrightarrow{\iota_C} & C+D & \xleftarrow{\iota_D} & D \\ & \searrow h\sharp\iota_E & \downarrow h+k & \swarrow k\sharp\iota_F & \\ & & E+F & &  \end{array} $ |
| fong-spivak-invitation_<br>DIA_0460 | 321  | —     | —            | —        |                                                                                                                                                                                                                                                                                                         |
| fong-spivak-invitation_<br>DIA_0461 | 321  | —     | —            | —        | $ \begin{array}{ccc} 1 & \longrightarrow & 1' \\ 2 & \longrightarrow & 2' \\ 3 & \longrightarrow & 3' \\ 4 & \longrightarrow & \\ 5 & \longrightarrow &  \end{array} $                                                                                                                                  |
| fong-spivak-invitation_<br>DIA_0462 | 321  | —     | —            | —        | $ \begin{array}{ccc} \emptyset & \xrightarrow{f} & X \\ g \downarrow & & \downarrow x \\ Y & \xrightarrow{y} & T \end{array} $                                                                                                                                                                          |
| fong-spivak-invitation_<br>DIA_0463 | 322  | —     | —            | —        | $ \begin{array}{ccc} & B & \longrightarrow Z \\ & \downarrow & \\ A & \longrightarrow & C \\ \downarrow & & \\ D & &  \end{array} $                                                                                                                                                                     |
| fong-spivak-invitation_<br>DIA_0464 | 322  | —     | —            | —        | $ \begin{array}{ccccc} & B & \longrightarrow & Z & \\ & \downarrow & & \downarrow & \\ A & \longrightarrow & Y & \xrightarrow{r} & R \\ \downarrow & & \downarrow & & \downarrow \\ X & \xrightarrow{r} & Q & \xrightarrow{r} & S \end{array} $                                                         |
| fong-spivak-invitation_<br>DIA_0465 | 322  | —     | —            | —        | $ \begin{array}{ccc} & B & \longrightarrow Z \\ & \downarrow & \\ A & \longrightarrow & Y \\ \downarrow & & \searrow \\ X & \longrightarrow & T \end{array} $                                                                                                                                           |

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| fong-spivak-invitation_<br>DIA_0466 | 322  | —     | —            | —        | <p><math>A + B \quad N + P \quad B + C</math></p>                                                                                                                                                                                                                                            |
| fong-spivak-invitation_<br>DIA_0467 | 323  | —     | —            | —        | <p><math>1 + 1 \quad 1 \quad 1 \quad \emptyset \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 + 1 \quad 1 \quad 1 \quad \emptyset</math></p> <p>Multiplication <math>\mu_1</math>      Unit <math>\eta_1</math>      Comultiplication <math>\delta_1</math>      Counit <math>\epsilon_1</math></p> |
| fong-spivak-invitation_<br>DIA_0468 | 323  | —     | —            | —        |                                                                                                                                                                                                                                                                                              |
| fong-spivak-invitation_<br>DIA_0469 | 323  | —     | —            | —        |                                                                                                                                                                                                                                                                                              |
| fong-spivak-invitation_<br>DIA_0470 | 324  | —     | —            | —        |                                                                                                                                                                                                                                                                                              |
| fong-spivak-invitation_<br>DIA_0471 | 324  | —     | —            | —        |                                                                                                                                                                                                                                                                                              |
| fong-spivak-invitation_<br>DIA_0472 | 325  | —     | —            | —        |                                                                                                                                                                                                                                                                                              |
| fong-spivak-invitation_<br>DIA_0473 | 325  | —     | —            | —        |                                                                                                                                                                                                                                                                                              |
| fong-spivak-invitation_<br>DIA_0474 | 326  | —     | —            | —        | <p><math>\eta \circ x = \emptyset</math></p> <p><math>= \emptyset</math></p>                                                                                                                                                                                                                 |

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| fong-spivak-invitation_<br>DIA_0475 | 326  | —     | —            | —        | $\eta \circ x \circ \epsilon = \emptyset$  $= \emptyset$                                                                      |
| fong-spivak-invitation_<br>DIA_0476 | 326  | —     | —            | —        | inner circles' ports<br>links<br>outer circle's ports                                                                                                                                                            |
| fong-spivak-invitation_<br>DIA_0477 | 326  | —     | —            | —        | inner circles' ports<br>links<br>outer circle's ports                                                                                                                                                            |
| fong-spivak-invitation_<br>DIA_0478 | 327  | —     | —            | —        | inner circles' ports<br>links<br>outer circle's ports                                                                                                                                                            |
| fong-spivak-invitation_<br>DIA_0479 | 327  | —     | —            | —        | $\begin{array}{ccccc} A & \xrightarrow{f} & B & \xrightarrow{g} & C \\ \downarrow h_1 & & \downarrow h_2 & \lrcorner & \downarrow h_3 \\ A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' \end{array}$                                                                                            |
| fong-spivak-invitation_<br>DIA_0480 | 327  | —     | —            | —        | $\begin{array}{ccccc} X & & & & \\ & \searrow p & & & \\ & & A & \xrightarrow{f} & B & \xrightarrow{g} & C \\ & \swarrow q & \downarrow h_1 & & \downarrow h_2 & \lrcorner & \downarrow h_3 \\ & & A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' \end{array}$                                  |
| fong-spivak-invitation_<br>DIA_0481 | 327  | —     | —            | —        | $\begin{array}{ccccc} X & & & & \\ & \searrow p & & & \\ & \searrow r & & & \\ & & B & \xrightarrow{g} & C \\ & \searrow q \circ f' & \downarrow h_2 & \lrcorner & \downarrow h_3 \\ & & B' & \xrightarrow{g'} & C' \end{array}$                                                                    |
| fong-spivak-invitation_<br>DIA_0482 | 327  | —     | —            | —        | $\begin{array}{ccccc} X & & & & \\ & \searrow p & & & \\ & \searrow r & & & \\ & \searrow q & & & \\ & & A & \xrightarrow{f} & B & \xrightarrow{g} & C \\ & & \downarrow h_1 & & \downarrow h_2 & \lrcorner & \downarrow h_3 \\ & & A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' \end{array}$ |

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| fong-spivak-invitation_<br>DIA_0483 | 328  | —     | —            | —        | $ \begin{array}{ccccc} X & & & & \\ \downarrow q & \searrow p & & & \\ A & \xrightarrow{f} & B & \xrightarrow{g} & C \\ \downarrow h_1 & & \downarrow h_2 & \lrcorner & \downarrow h_3 \\ A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' \end{array} $                               |
| fong-spivak-invitation_<br>DIA_0484 | 328  | —     | —            | —        | $ \begin{array}{ccccc} X & & & & \\ \downarrow q & \searrow r & & & \\ A & \xrightarrow{f} & B & \xrightarrow{g} & C \\ \downarrow h_1 & & \downarrow h_2 & \lrcorner & \downarrow h_3 \\ A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' \end{array} $                               |
| fong-spivak-invitation_<br>DIA_0485 | 328  | —     | —            | —        | $ \begin{array}{ccccc} X & & & & \\ \downarrow q & \searrow p & & & \\ A & \xrightarrow{f} & B & \xrightarrow{g} & C \\ \downarrow h_1 & & \downarrow h_2 & \lrcorner & \downarrow h_3 \\ A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' \end{array} $                               |
| fong-spivak-invitation_<br>DIA_0486 | 328  | —     | —            | —        | $ \begin{array}{ccccc} X & & & & \\ \downarrow g_1 & \searrow g_2 & & & \\ A & \xrightarrow{\text{id}_A} & A & & \\ \downarrow \text{id}_A & \lrcorner & \downarrow f & & \\ A & \xrightarrow{f} & B & &  \end{array} $                                                                  |
| fong-spivak-invitation_<br>DIA_0487 | 329  | —     | —            | —        | $ \begin{array}{ccc} A' & \xrightarrow{f'} & B' \\ i' \downarrow & \lrcorner & \cong \downarrow i \\ A & \xrightarrow{f} & B \end{array} $                                                                                                                                               |
| fong-spivak-invitation_<br>DIA_0488 | 329  | —     | —            | —        | $ \begin{array}{ccc} A & \xrightarrow{f} & B \\ \parallel \lrcorner & & \parallel \\ A & \xrightarrow{f} & B \end{array} $                                                                                                                                                               |
| fong-spivak-invitation_<br>DIA_0489 | 329  | —     | —            | —        | $ \begin{array}{ccccc} & & A & & \\ & & \parallel & & \\ A' & \xrightarrow{g} & A & & \\ \downarrow f' & \lrcorner & \downarrow f & & \\ A' & \xrightarrow{f'} & A & & \\ \downarrow f' & \lrcorner & \downarrow f & & \\ A' & \xrightarrow{f'} & B' & \xrightarrow{h} & B \end{array} $ |
| fong-spivak-invitation_<br>DIA_0490 | 329  | —     | —            | —        | $ \begin{array}{ccccc} & & A & & \\ & & \parallel & & \\ A' & \xrightarrow{g} & A & & \\ \downarrow f' & \lrcorner & \downarrow f & & \\ A' & \xrightarrow{f'} & A & & \\ \downarrow f' & \lrcorner & \downarrow f & & \\ A' & \xrightarrow{f'} & B' & \xrightarrow{h} & B \end{array} $ |

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| fong-spivak-invitation_<br>DIA_0491 | 330  | —     | —            | —        |                                                                                                                                                                                                                                                                                                                                                                                                |
| fong-spivak-invitation_<br>DIA_0492 | 330  | —     | —            | —        | $  \begin{array}{ccccccc}  (0,0) & \leftarrow & (0,1) & \leftarrow & (0,2) & \leftarrow & (0,3) \cdots \\  \uparrow & & \uparrow & & \uparrow & & \\  (1,0) & \leftarrow & (1,1) & \leftarrow & (1,2) & \cdots & \\  \uparrow & & \uparrow & & & & \\  (2,0) & \leftarrow & (1,1) & \cdots & & & \\  \uparrow & & & & & & \\  (3,0) & \cdots & & & & & \\  \vdots & & & & & &   \end{array}  $ |
| fong-spivak-invitation_<br>DIA_0493 | 333  | —     | —            | —        |                                                                                                                                                                                                                                                                                                                                                                                                |
| fong-spivak-invitation_<br>DIA_0494 | 333  | —     | —            | —        | $  \begin{array}{c}  X := \begin{array}{ccccccc} & b_3 & & & & & \\ a_2 & & b_2 & & & & e_2 \\ \bullet & & \bullet & & & & \bullet \\ a_1 & & b_1 & & c_1 & & e_1 \\ \bullet & & \bullet & & \bullet & & \bullet \end{array} \\  \downarrow f \\  Y := \begin{array}{cccccc} a & b & c & d & e \\ \bullet & \bullet & \bullet & \bullet & \bullet \end{array}  \end{array}  $                  |
| fong-spivak-invitation_<br>DIA_0495 | 333  | —     | —            | —        | $  \begin{array}{c}  X := \begin{array}{cccccc} a_2 & & c_1 & & & e_2 \\ \bullet & & \bullet & & & \bullet \\ e_1 & & b_1 & & b_2 & b_3 & a_1 \\ \bullet & & \bullet & & \bullet & \bullet & \bullet \end{array} \\  \downarrow f \\  Y := \begin{array}{cccccc} a & b & c & d & e \\ \bullet & \bullet & \bullet & \bullet & \bullet \end{array}  \end{array}  $                              |
| fong-spivak-invitation_<br>DIA_0496 | 334  | —     | —            | —        |                                                                                                                                                                                                                                                                                                                                                                                                |

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| fong-spivak-invitation_<br>DIA_0497 | 334  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0498 | 334  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0499 | 334  | —     | —            | —        |       |
| fong-spivak-invitation_<br>DIA_0500 | 335  | —     | —            | —        |       |
| fong-spivak-invitation_<br>PIC_0001 | 13   | —     | —            | —        |       |
| fong-spivak-invitation_<br>PIC_0002 | 13   | —     | —            | —        |       |

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| fong-spivak-invitation_PIC_0003 | 296  | —     | —            | —        |       |
| fong-spivak-invitation_PIC_0004 | 296  | —     | —            | —        |       |
| fong-spivak-invitation_PIC_0005 | 296  | —     | —            | —        |       |